Exercise 3

Part 1

4)

$$\frac{E}{m} = \left(1 - \frac{2M}{r}\right)^{\frac{1}{2}} \gamma_{shell}$$

Vi bruker betingelser ved $t=0$.

$$M = (2.8113 \cdot 10^{7})\, m_{\odot} = 4.15 \cdot 10^{10}\ m$$
$$r = 1\,AU = 1.50 \cdot 10^{11}\ m$$
$$v = 0.214c = 0.214$$

Vi får:

$$\frac{E}{m} = 0.699096 \qquad (\text{enhetsløs})$$

5)

$$\frac{E}{m} = \left(1 - \frac{2M}{r}\right) \frac{\Delta t}{\Delta \tau}$$

$$\Downarrow$$

$$\Delta\tau \frac{E}{m} = \left(1 - \frac{2M}{r}\right) \Delta t$$

$$\Downarrow$$

$$\Delta\tau = \frac{1 - \frac{2M}{r}}{E/m} \Delta t$$

6)

$$\Delta\tau = \frac{1 - \frac{2M}{r}}{E/m} \frac{\Delta t_{shell}}{\sqrt{1 - \frac{2M}{r_0}}}$$

$$\Delta\tau = \frac{1 - \frac{2M}{r}}{E/m\left(1 - \frac{2M}{r_0}\right)^{\frac{1}{2}}} \Delta t_{shell}$$

7)

Vi velger Planet-systemet (shell)

$$\Delta\tau = 20.36\,s = 6.104 \cdot 10^{9}\ m$$

$$t_1 = 20.0138, \quad t_2 = 43.3633$$
$$\Delta t_1 = 23.35\,s = 7.0 \cdot 10^{9}\ m$$

$$t_{n-1} = 1020.71\,s \quad t_n = 1444.33\,s$$
$$\Delta t_n = 423.62\,s = 1.27 \cdot 10^{11}$$

Vi har:

$$\Delta\tau = \frac{1 - \frac{2M}{r}}{E/m\left(1 - \frac{2M}{r_0}\right)^{\frac{1}{2}}} \Delta t_{shell}$$

$$\frac{\Delta\tau}{\Delta t_{shell}} \cdot \frac{E}{m} \left(1 - \frac{2M}{r_0}\right)^{\frac{1}{2}} = 1 - \frac{2M}{r}$$

$$r = 2M\left(1 - \frac{\Delta\tau}{\Delta t_{shell}} \cdot \frac{E}{m}\left(1 - \frac{2M}{r_0}\right)^{\frac{1}{2}}\right)^{-1}$$

Vi setter inn og får

$$r_1 = 0.9355\ AU$$
$$= 1.686\ r_C \qquad \leftarrow \text{Schwarzschild-radius}$$

$$r_2 = 0.5677\ AU$$
$$= 1.0229\ r_C$$

Part 2

4)

$$\frac{E}{m} = \left(1 - \frac{2M}{r}\right) \frac{\Delta t}{\Delta \tau}$$

$$v_r = \frac{dr}{dt} = \frac{dr}{d\tau} \cdot \frac{d\tau}{dt}$$

$$= \frac{dr}{d\tau} \cdot \frac{\left(1 - \frac{2M}{r_s}\right)}{E/m}$$

$$\Downarrow$$

$$\frac{dr}{d\tau} \frac{\left(1 - \frac{2M}{r_s}\right)}{E/m} = -\left(1 - \frac{2M}{r_\gamma}\right)$$

$$\Downarrow$$

$$\Delta r = -\frac{1 - \frac{2M}{r_\gamma}}{1 - \frac{2M}{r_s}} \frac{E}{m} \Delta\tau$$

Exercise 5

2)

Vi har

$$\frac{E}{m} = \left(1 - \frac{2M}{r}\right)\frac{dt}{d\tau}$$

og

$$dt = \frac{dt_{shell}}{\left(1 - \frac{2M}{r}\right)^{\frac{1}{2}}}$$

dersom vi antar et veldig kort tidsintervall. Vi får

$$\frac{E}{m} = \left(1 - \frac{2M}{r}\right)^{1/2} \frac{dt_{shell}}{d\tau}.$$

Siden vi har antatt kort tidsintervall får vi

$$\frac{dt_{shell}}{d\tau} = \frac{1}{\sqrt{1 - v_{shell}^2}} = \gamma_{shell}$$

$$\Downarrow$$

$$\frac{E}{m} = \left(1 - \frac{2M}{r}\right)^{1/2} \gamma_{shell}$$

3)

$$V_{eff}(r) = \Bigg(\underbrace{\left(1 - \frac{2M}{r}\right)\left(1 + \frac{(L/m)^2}{r^2}\right)}_{d}\Bigg)^{1/2}$$

Vi deriverer:

$$V_{eff}'(r) = \left(\frac{1}{2} d^{-\frac{1}{2}}\right)\left(\frac{2M}{r^2}\left(1 + \frac{(L/m)^2}{r^2}\right) + \left(1 - \frac{2M}{r}\right)\frac{2(L/m)^2}{r^3}\right)$$

$$= \left(\frac{1}{2} d^{-\frac{1}{2}}\right)\left(\frac{2M}{r^2} + \frac{2M(L/m)^2}{r^4} - \frac{2(L/m)^2}{r^3} + \frac{4M(L/m)^2}{r^4}\right)$$

Vi setter det lik 0:

$$\left(\frac{1}{2} d^{-\frac{1}{2}}\right)\left(\frac{2M}{r^2} + \frac{2M(L/m)^2}{r^4} - \frac{2(L/m)^2}{r^3} + \frac{4M(L/m)^2}{r^4}\right) = 0$$

$d = 0$ kun dersom $r = 2M$. Vi ganger den vekk, og husker at dette ikke kan være en løsning

$$\frac{2M}{r^2} + \frac{2M(L/m)^2}{r^4} - \frac{2(L/m)^2}{r^3} + \frac{4M(L/m)^2}{r^4} = 0$$

$$\Downarrow$$

$$\frac{1}{r^2} - \frac{(L/m)^2}{Mr^3} + \frac{3(L/m)^2}{r^4} = 0$$

$$\Downarrow$$

$$r^2 - \frac{(L/m)^2}{M} r + 3(L/m)^2 = 0$$

Vi løser med ABC:

$$r = \frac{\frac{(L/m)^2}{M} \pm \left(\frac{(L/m)^4}{M^2} - 12(L/m)^2\right)^{\frac{1}{2}}}{2}$$

$$\Downarrow$$

$$r = \frac{(L/m)^2}{2M}\left(1 \pm \left(1 - \frac{12M}{(L/m)^2}\right)^{\frac{1}{2}}\right)$$

Vi vet at dette toppunktet vil være ved en lavere r enn bunnpunktet, så vi velger minus:

$$r_{crit} = \frac{(L/m)^2}{2M}\left(1 - \left(1 - \frac{12M}{(L/m)^2}\right)^{\frac{1}{2}}\right)$$

4)

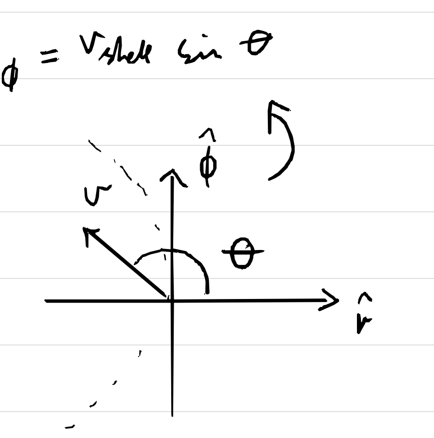
Vi har:

$$\frac{L}{m} = r^2 \frac{d\phi}{d\tau} = r^2 \frac{d\phi}{dt_{shell}} \frac{dt_{shell}}{d\tau}$$

$$= \underbrace{r\frac{d\phi}{dt_{shell}}}_{v_{shell}} \cdot \underbrace{r}_{r} \cdot \underbrace{\frac{dt_{shell}}{d\tau}}_{\gamma_{shell}}$$

$$= v_{shell_\phi}\, r\, \gamma_{shell}$$

Vi har $R = r$ og $v_{shell_\phi} = v_{shell} \sin\theta$



Vi får:

$$\frac{L}{m} = R\gamma_{shell} v_{shell} \sin\theta$$

6)

Vi har

$$\left(1 - \frac{2M}{r}\right)\frac{dt}{d\tau} = (E/m)$$

$$\Downarrow$$

$$\left(\frac{\left(1 - \frac{2M}{r}\right)}{(E/m)}\right)^2 = \frac{d\tau^2}{dt^2}$$

Vi har også at

$$\Delta\tau^2 = \Delta s^2 = \left(1 - \frac{2M}{r}\right)\Delta t^2 - \frac{\Delta r^2}{\left(1 - \frac{2M}{r}\right)}$$

Vi løser:

$$\left(\frac{\left(1 - \frac{2M}{r}\right)}{(E/m)}\right)^2 = \frac{\left(1 - \frac{2M}{r}\right)\cancel{\Delta t^2} - \frac{\Delta r^2}{\left(1 - \frac{2M}{r}\right)}}{\cancel{\Delta t^2}}$$

$$\frac{\left(1 - \frac{2M}{r}\right)^2}{(E/m)^2} = \left(1 - \frac{2M}{r}\right) - \frac{1}{\left(1 - \frac{2M}{r}\right)}\frac{dr^2}{dt^2}$$

$$\frac{dr^2}{dt^2} \cdot \frac{1}{\left(1 - \frac{2M}{r}\right)} = \left(1 - \frac{2M}{r}\right) - \frac{\left(1 - \frac{2M}{r}\right)^2}{(E/m)^2}$$

$$\frac{dr^2}{dt^2} = \left(1 - \frac{2M}{r}\right)^2 - \frac{\left(1 - \frac{2M}{r}\right)^3}{(E/m)^2}$$

$$\frac{dr^2}{dt^2} = \left(1 - \frac{2M}{r}\right)^2\left(1 - \frac{1 - \frac{2M}{r}}{(E/m)^2}\right)$$

$$\frac{dr}{dt} = -\left(1 - \frac{2M}{r}\right)\left(1 - \frac{1 - \frac{2M}{r}}{(E/m)^2}\right)^{\frac{1}{2}}$$

Vi velger − siden r faller i

Vi ønsker å finne $\frac{dr}{d\tau}$:

$$\frac{dr}{d\tau} = \frac{dr}{dt} \cdot \frac{dt}{d\tau}$$

$$= -\left(1 - \frac{2M}{r}\right)\left(1 - \frac{1 - \frac{2M}{r}}{(E/m)^2}\right)^{\frac{1}{2}} \cdot \frac{(E/m)}{\left(1 - \frac{2M}{r}\right)}$$

$$= -\left(1 - \frac{1 - \frac{2M}{r}}{(E/m)^2}\right)^{\frac{1}{2}}(E/m)$$

$$= \left((E/m)^2 - 1 + 2\frac{M}{r}\right)^{\frac{1}{2}}$$

Merk at dette forenkles til $-\sqrt{\frac{2M}{r}}$ dersom $(E/m) = 1$.

$$\Downarrow$$

$$d\tau = \left((E/m)^2 - 1 + \frac{2M}{r}\right)^{-\frac{1}{2}} dr$$

Vi legger inn i en integral-kalkulator og får:

$$\tau = 5.01\ s$$